


INFORMATICS INSTITUTE OF TECHNOLOGY

**Foundation certificate in higher education.**

**Module:** DOC 313 Introduction to Programming I

**Module Leader**: Ms. Sudarshan Welihinda

**Type of Assignment**: Individual Coursework

**Submission Date**: 05/04/2022

**Name**: Venuri Jayasinghe

**Student ID**: 20211228

# Abstract

This report consists of one programming question answered methodically.

The cords are written in python language.

# I. Acknowledgment

Firstly, I am grateful to our module lecturer Mr. Sudharshana Welihinda, our tutorial lecturer Ms. Tharushi Amarasinghe, and Ms. Salitha Dinushika for their leadership and assistance throughout the report. Thank you for the continuous guidance rendered throughout this project.

Along with that, I am very grateful to the Informatics Institute of Technology for providing us with the study materials that we required for this report.

Finally, I would like to thank my parents who were always with me motivating and inspiring me in all situations.

# II. Table Of Contents

## Contents

**Abstract** ..... ii

**Acknowledgment** ..... iii

**Table Of Contents** ..... iv

**List of figures** ..... v

**List of tables** ..... v

1. **Problem understanding** ..... 1
2. **Algorithm** ..... 2
3. **Python codes** ..... 5
4. **Test cases** ..... 7
5. **Screenshots of the test cases** ..... 8

# III. List of figures

Figure 1. Python code (part 1) ..... 5
Figure 2 Python codes ( part 2) ..... 5
Figure 3. Python codes (part 3) ..... 6
Figure 4. Python codes (part 4) ..... 6
Figure 5. Test Case 1 ..... 8
Figure 6 Test Case 2 ..... 8
Figure 7 Test case 3 ..... 8
Figure 8. Test Case 04 ..... 9
Figure 9. Test Case 05 ..... 9
Figure 10. Test Case 06 ..... 9
Figure 11. Test Case 07 ..... 9
Figure 12. Test Case 8 ..... 10
Figure 13. Test Case 9 ..... 10
Figure 14. Test case 10 ..... 10

# IV. List of tables

Table 1 Test cases ..... 7

# 1. Problem understanding

It is required to develop a python program to develop the shapes calculator. This program should calculate the surface area, volume, and base area of a cone. Also, the volume and surface area of the rectangular pyramid needs to be calculated. The program should ask for measurements required to calculate the surface area, volume, and base area for a selected shape. Users can input the measurements as decimals, but those values should be positive. If they input an invalid value (negative value), the program should display an error message.

When an option is selected from the menu by the user, the program should ask for necessary inputs to calculate and display the respective value. Once the result is displayed, the program needs to ask whether the user wants to continue or not with another shape. If the user wants to continue, he/she should be able to do so without exiting the program. If so, the program should repeat again. But, if the user does not want to continue, the program should end. Also, if the user selects the exit option the program should terminate.

The data type of inputs should be floats, as the user can input decimals values as well. All the formulas given were used in the program.

## 2. Algorithm

Step 1: Start

Step 2: Initialize the variables

Step 3: Get the user to input the preferred option

Step 4: If the preferred option is the first option get the value of the radius(r) and slant height (H)of the cone. If those values are negative, display an error message.

Step 5: Repeat step 4 until the user inputs positive values. And then calculate the surface area of a cone. Surface_ area =(3.14*r*r) +(3.14*r*H)

Step 6: Display the Surface_ area.

Step 7: Get the opinion of the user to continue with another shape or not. If the user enters “Y” then repeat step 3.

Step 8: If the preferred option is the second option, get the values of the radius(r) and height(H) of the cone. If the values are negative display an error message.

Step 9: repeat step 8 till the user enters positive values. And then calculate the volume of the cone. (Volume= 1/3*3.14*r*r*H)

Step 10: Display the volume.

Step 11: repeat step 7

Step 12: If the preferred option is the third option, get the value of the radius (r) of the cone If those values are negative, display an error message.

Step 13: Repeat step 12 until the user inputs positive values. And then calculate the base area of a cone. (Base area= 3.14*r*r)

Step 14: Display the base area..

Step 15: Repeat step 7

Step 16: If the required option is the fourth option, get the length(a), width(b), and height(H) of the rectangular-based pyramid. If the values are negative display an error message.

Step 17: Repeat step 16 until the user inputs positive values. And then calculate the Volume of the pyramid. (volume=a*b*H/3).

Step 18: Display the Volume of the rectangular-based pyramid.

Step 19: Repeat step 7

Step 20: If the required option is the fifth option, get the length(a), width(b), and height(H) of the rectangular-based pyramid. If the values are negative display an error message.

Step 21: Repeat step 20 until the user inputs positive values. And then calculate the surface area of the rectangular based pyramid.
{area=((a*b)+(a*(((b/2)**2)+(H**2))**0.5)+(b*(((a/2)**2)+(H**2))**5)) }

Step 22: Display the surface area of a rectangular based pyramid.

Step 23: Repeat step 7

Step 24: If the required option is the sixth option, Display “Thank You” and then Break the option.

Step 25: Stop.

# 3. Python codes

```
l=0
a=0
b=0
H=0
surface_area=0
volume=0
base_area=0
option=0
opinion=str("Y")
#main menu
while (opinion=="Y"):
    print()
    print("SHAPES CALCULATOR")
    print()
    print()
    print("1. Calculate Surface area of a cone ")
    print("2. Calculate Volume of a cone ")
    print("3. Calculate Base Area of a cone ")
    print("4. Calculate Volume of a rectangular pyramid ")
    print("5. Calculate Surface area of a rectangular pyramid ")
    print("6. Exit ")
    print()
    print()
    option=int(input("Enter preffered option:"))
   #check for the condition,
    if (option==1):
        r=float(input("Enter the radius of the cone in centimeters:"))
        while r<0:
           print("Negative numbers are not allowed ")
           r=float(input("Enter a postive number and try again:"))

        l=float(input("Enter the slant height of the cone in centimeters:"))
        while l<0:
           print("Negative numbers are not allowed ")
           l=float(input("Enter a positive value and try again:"))
         #calculating the surface area of a cone
        surface_area= (3.14*r*r)+(3.14*r*l)
        print( "surface area of a cone :",round(surface_area,2),"sq.cm")
        opinion=str(input("Do you want to continue (Y/N):"))
```

*Figure 1. Python code (part 1)*

```
    else:
        if (option==2):
            #Inputs required to calculate the volume of a cone
            r=float(input("Enter the radius of the cone in centimeters:"))
            while r<0:
                print("Negative numbers are not allowed")
                r=float(input("Enter a positive number :"))
            H=float(input("Enter the height of the cone in centimeters:"))
            while H<0:
               print("Negative numbers are not allowed ")
               H=float(input("Enter a positive value"))
             #calculating the volume of a cone
            Volume=(1/3*3.14*r*r*H)
            print("The volume of a cone is :",round(Volume,2),"cubic cm")
            opinion=str(input("Do you want to continue (Y/N):"))
        else:
            if (option==3):
               #Inputs required to calculate the base area of a cone
               r=float(input("Enter the radius of the cone in centimeters:"))
               while r<0:
                  print("Negative numbers are not allowed")
                  r=float(input("Enter a positive number :"))
               base_area= 3.14*r*r
               #calculating the base area of a cone
               print("The Base area of the cone is :",round(base_area,2), "sq.cm")
               opinion=str(input("Do you want to continue (Y/N):"))
```

*Figure 2 Python codes ( part 2)*

```
else:
    if (option==4):
        #Inputs required to calculate the volume of a rectangular pyramid
        a=float(input("Enter the base length of the rectangular pyramid in centimeters:"))
        while a<0:
            print("Negative numbers are not allowed")
            a=float(input("Enter a positive number :"))
        b=float(input("Enter the base width of the rectangular pyramid in centimeters:"))
        while b<0:
            print("Negative numbers are not allowed")
            b=float(input("Enter a positive number :"))
        H=float(input("Enter the height of the rectangular base pyramid:"))
        while H<0:
            print("Negative numbers are not allowed")
            H=float(input("Enter a positive number :"))
        #calculating the volume of rectangular pyramid
        Volume=a*b*H/3
        print("The volume of the rectangular pramid is:",round(Volume,2), "cubic.cm")
        opinion=str(input("Do you want to continue (Y/N):"))
```

*Figure 3. Python codes (part 3)*

```
        else:
            if (option==5):
            #Inputs required to calculate the surface area of a rectangular pyramid
                a=float(input("Enter the base length of the rectangular pyramid in centimeters:"))
                while a<0:
                    print("Negative numbers are not allowed")
                    a=float(input("Enter a positive number :"))
                b=float(input("Enter the base width of the rectangular pyramid in centimeters:"))
                while b<0:
                    print("Negative numbers are not allowed")
                    b=float(input("Enter a positive number :"))
                H=float(input("Enter the height of the rectangular base pyramid in centimneters:"))
                while H<0:
                    print("Negative numbers are not allowed")
                    H=float(input("Enter a positive number :"))
            #calculating the surface area of the rectangular pyramid
                area=((a*b)+(a*(((b/2)**2)+(H**2))**0.5)+(b*(((a/2)**2)+(H**2))**5))
                print("The surface area of the rectangular pramid is:",round(area,2), "sq.cm")
                opinion=str(input("Do you want to continue (Y/N):"))
            else:
            #program terminate
                if (option==6):
                    print("Thank You")
                    break
```

*Figure 4. Python codes (part 4)*

# 04. Test cases

*Table 1 Test cases*

| Test case # | Inputs | Expected output | Actual output | Remarks |
|---|---|---|---|---|
| 1 | Option=1<br>r= 18 cm<br>l= 15.5cm | surface area of a cone: 1893.42 sq.cm<br>Do you want to continue (Y/N): | surface area of a cone: 1893.42 sq.cm<br>Do you want to continue (Y/N): | Test Case Pass |
| 2 | Option=2<br>r=12.25 cm<br>H=14.5 cm | The volume of a cone is: 2277.45 cubic cm<br>Do you want to continue (Y/N): | The volume of a cone is: 2277.45 cubic cm<br>Do you want to continue (Y/N): | Test Case Pass |
| 3 | Option=3<br>r=16.25 cm | The Base area of the cone is : 829.16 sq. cm<br>Do you want to continue (Y/N): | The Base area of the cone is : 829.16 sq.cm<br>Do you want to continue (Y/N): | Test Case Pass |
| 4 | Option=4<br>a=14.32 cm<br>b=10 cm<br>H=12.5 cm | The volume of the rectangular pyramid is: 584.73 cubic.cm<br>Do you want to continue (Y/N): | The volume of the rectangular pyramid is: 584.73 cubic.cm<br>Do you want to continue (Y/N): | Test Case Pass |
| 5 | Option=5<br>a=12.5 cm<br>b=26.7 cm<br>H=14.2 cm | The surface area of the rectangular pyramid is: 991.62 sq. cm<br>Do you want to continue (Y/N): | The surface area of the rectangular pyramid is: 991.62 sq. cm<br>Do you want to continue (Y/N): | Test Case Pass |
| 6 | Option=6 | Thank You | Thank You | Test Case Pass |
| 7 | Option=2<br>r= -12.2 cm<br>H=10.2 cm | Negative numbers are not allowed<br>Enter a positive number : | Negative numbers are not allowed<br>Enter a positive number : | Test Case Pass |
| 8 | Option =4<br>a=7.5 cm<br>b=12.2 cm<br>H= -10 cm | Negative numbers are not allowed<br>Enter a positive number : | Negative numbers are not allowed<br>Enter a positive number : | Test Case Pass |
| 9 | Option=3<br>r= -52.2 cm | Negative numbers are not allowed<br>Enter a positive number | Negative numbers are not allowed<br>Enter a positive number | Test Case Pass |
| 10. | Option=1<br>r= -15 cm<br>l= 15.5cm | Negative numbers are not allowed<br>Enter a positive number | Negative numbers are not allowed<br>Enter a positive number | Test Case Pass |

# 05. Screenshots of the test cases

01.

```
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

==================== RESTART: C:\Users\pc\Desktop\cw prgg.py ====================

SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:1
Enter the radius of the cone in centimeters:18
Enter the slant height of the cone in centimeters:15.5
surface area of a cone : 1893.42 sq.cm
Do you want to continue (Y/N):
```

*Figure 5. Test Case 1*

02.

```
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.

==================== RESTART: C:\Users\pc\Desktop\cw prgg.py ====================

SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:2
Enter the radius of the cone in centimeters:12.25
Enter the height of the cone in centimeters:14.5
The volume of a cone is : 2277.45 cubic cm
Do you want to continue (Y/N):
```

*Figure 6 Test Case 2*

03.

```


Enter preffered option:3
Enter the radius of the cone in centimeters:16.25
The Base area of the cone is : 829.16 sq.cm
Do you want to continue (Y/N):Y

SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit
```

*Figure 7 Test case 3*

04.

```
SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:4
Enter the base length of the rectangular pyramid in centimeters:14.32
Enter the base width of the rectangular pyramid in centimeters:10
Enter the height of the rectangular base pyramid:12.5
The volume of the rectangular pyramid is: 596.67 cubic.cm
Do you want to continue (Y/N):Y
```

*Figure 8. Test Case 04*

05.

```
SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:5
Enter the base length of the rectangular pyramid in centimeters:12.5
Enter the base width of the rectangular pyramid in centimeters:26.7
Enter the height of the rectangular base pyramid in centimneters:14.2
The surface area of the rectangular pramid is: 991.62 sq.cm
Do you want to continue (Y/N):
```

*Figure 9. Test Case 05*

06.

```
SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:6
Thank You
```

*Figure 10. Test Case 06*

07.

```
SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:2
Enter the radius of the cone in centimeters:-12.2
Negative numbers are not allowed
```

*Figure 11. Test Case 07*

08.

```
SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:4
Enter the base length of the rectangular pyramid in centimeters:7.5
Enter the base width of the rectangular pyramid in centimeters:12.2
Enter the height of the rectangular base pyramid:-10
Negative numbers are not allowed
```

*Figure 12. Test Case 8*

09.

```
SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:1
Enter the radius of the cone in centimeters:-15
Negative numbers are not allowed
```

*Figure 13. Test Case 9*

10.

```
SHAPES CALCULATOR


1. Calculate Surface area of a cone
2. Calculate Volume of a cone
3. Calculate Base Area of a cone
4. Calculate Volume of a rectangular pyramid
5. Calculate Surface area of a rectangular pyramid
6. Exit


Enter preffered option:1
Enter the radius of the cone in centimeters:-15
Negative numbers are not allowed
```

*Figure 14. Test case 10*